11.2.1 Computing the SVD from the Spectral decomposition

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Week 11 due Apr 5, 2025 12:56 IST

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matrix A? And, you know, we notice, hey, we know what V is. Now when we compute

the Spectral Decomposition it is not necessarily the case that the

eigenvalues come out from largest to smallest. So we may have to put a

permutation on our matrix Sigma in order to make it have its entries ordered from

largest to smallest. But, once we have that, we can merely take

the square root of the diagonal elements.

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Reading assignment

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Read Unit 11.2.1 of the notes. [LINK]

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? In Homework 11.2.1.2
Why D has real positive values if A is nonsingular, so is A^H A? We don't kno... 4

Homework 11.2.1.1

Homework 11.2.1.1

1/1 point (graded)

Let $A \in \mathbb{C}^{m \times m}$ be nonsingular and $A^H A = QDQ^H$, the Spectral Decomposition of $A^H A$. Give a formula for U, V, and Σ so that $A = U \Sigma V^H$ is the SVD of A. (Notice that A is square.)

- [x] done

✔

Solution

For a complete solution, see the notes .

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ℹ Answers are displayed within the problem

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dictates how that happens. Now it turns out that the condition number of the matrix also impacts how relative error in the matrix gets amplified into the accuracy of the eigenvalues and eigenvectors that are computed. So that's a conditioning problem. The other problem is: Is the algorithm that we're executing numerically stable? Now it turns out that's when you form A Hermitian A, well we already saw this. That squares the condition number

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Homework 11.2.1.2

1/1 point (graded)

Let $A \in \mathbb{C}^{m \times n}$ have full column rank and let $A^H A = QDQ^H$, the Spectral Decomposition of $A^H A$. Give a formula for the Reduced SVD of A.

- [x] done

✔

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Homework 11.2.1.3

1/1 point (graded)

Let $A \in \mathbb{C}^{m \times n}$ be of rank r and

$$A^H A = \left(\begin{array}{c|c} Q_L & Q_R \end{array} \right) \left(\begin{array}{c|c} D_{TL} & 0_{r \times (n-r)} \\ \hline 0_{(n-r) \times r} & 0_{(n-r) \times (n-r)} \end{array} \right) \left(\begin{array}{c|c} Q_L & Q_R \end{array} \right)^H$$

be the Spectral Decomposition of $A^H A$, where $Q_L \in \mathbb{C}^{n \times r}$ and, for simplicity, we assume the diagonal elements of D_{TL} are ordered from largest to smallest. Give a formula for the Reduced SVD of A.

- [x] done

Solution

For a complete solution, see the notes .

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